

Fixed-Point Theory and Operator Algebra for a Clipped-Linear Generative Pipeline

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Abstract

We study a composite operator pipeline $G = U \blacksquare F \blacksquare K \blacksquare C$ acting on real-valued states, where each operator is defined by an explicit pointwise formula. We prove: (1) existence and uniqueness of fixed points for the one-dimensional clipped-linear map g_b , with global convergence at geometric rate $1/2$; (2) contraction of a two-dimensional coupled extension for coupling strength $|\alpha| < 1/2$, with a unique globally attracting fixed point; (3) a saturated pitchfork bifurcation at the critical coupling $|\alpha| = 1/2$, established via centre-manifold reduction; and (4) an algebraic separation theorem showing that no matrix of dimension $n < 33$ with spectral radius 1 can produce trace 33 at any power. A Lean 4 formal nucleus verifies the core fixed-point theorem. The numerical behaviour of discrete grid implementations — including any specific persistent-homology invariants — is not claimed here and remains an open question.

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1. Operator Definitions

We define four operators on real-valued functions $u : \mathbb{Z}^2 \rightarrow \mathbb{R}$ (or on $\mathbb{R}$ in the one-dimensional setting). Every operator is given by an explicit pointwise formula; nothing is left implicit.

Definition 1.1 (Comp, C).

$C(u)(x,y) = \text{clip}(\max(0, \text{mean}_{3 \times 3}(u)(x,y) - \tau), 0, 1)$, where $\text{mean}_{3 \times 3}(u)(x,y) = (1/9) \sum_{i,j \in \{-1,0,1\}} u(x+i, y+j)$ and $\tau \in \mathbb{R}$ is a fixed threshold parameter.

Definition 1.2 (Konst, K).

$K(u)(x,y) = \text{clip}(0.5 \cdot u(x,y) + b, -1, 1)$, where $b \in \mathbb{R}$ is a fixed bias parameter.

Definition 1.3 (Fold, F).

$F(u)(x,y) = (u(x,y) + u(x-1, y))/2$, with wrap-around boundary conditions.

Definition 1.4 (Unfold, U).

$U(u)(x,y) = u(x \bmod n, y \bmod n)$ on the $2n \times 2n$ tiling of u , cropped to $n \times n$.

Definition 1.5 (Composite operator).

$$G = U \blacksquare F \blacksquare K \blacksquare C.$$

In the one-dimensional setting the relevant operator is the scalar map $g_b(x) = 0.5 \cdot \kappa(x) + b$, where $\kappa(x) = \max(-1, \min(1, x))$ is the clipping function. This is the Konst operator restricted to one dimension, and is the object studied in Sections 2 and 3.

2. One-Dimensional Fixed-Point Theory**Theorem 2.1 (Existence, uniqueness, and explicit form).**

The map g_b has a unique fixed point $x^*(b)$, given by:

$$x^*(b) = b - 1/2 \quad \text{if } b < -1/2;$$

$$x^*(b) = 2b \quad \text{if } |b| \leq 1/2;$$

$$x^*(b) = b + 1/2 \quad \text{if } b > 1/2.$$

Proof. We solve $g_b(x) = x$ by cases on the value of $\kappa(x)$. **Case 1:** $x \leq -1$. Then $\kappa(x) = -1$ and $0.5(-1) + b = x$ gives $x = b - 1/2$. This satisfies $x \leq -1$ iff $b \leq -1/2$. **Case 2:** $-1 \leq x \leq 1$. Then $\kappa(x) = x$ and $0.5x + b = x$ gives $x = 2b$. This satisfies $|x| \leq 1$ iff $|b| \leq 1/2$. **Case 3:** $x \geq 1$. Then $\kappa(x) = 1$ and $0.5(1) + b = x$ gives $x = b + 1/2$. This satisfies $x \geq 1$ iff $b \geq 1/2$. The three cases partition $\mathbb{R}$ and each yields exactly one fixed point, establishing uniqueness. ■

Lemma 2.2 (Lipschitz contraction).

g_b is a contraction on $\mathbb{R}$ with Lipschitz constant $L = 1/2 < 1$, uniformly in b .

Proof. For any $x, y \in \mathbb{R}$: $|g_b(x) - g_b(y)| = (1/2)|\kappa(x) - \kappa(y)| \leq (1/2)|x - y|$, since κ is 1-Lipschitz (its derivative is 0 or 1 a.e.). ■

Theorem 2.3 (Global convergence).

For any initial $x_0 \in \mathbb{R}$, the orbit $x_{t+1} = g_b(x_t)$ satisfies $|x_t - x^*(b)| \leq (1/2)^t |x_0 - x^*(b)|$ for all $t \geq 0$.

Proof. $\mathbb{R}$ with the standard metric is a complete metric space. By Lemma 2.2, g_b is a contraction with constant $1/2$. The Banach fixed-point theorem gives a unique fixed point $x^*(b)$ and the geometric convergence estimate $|x_t - x^*(b)| \leq (1/2)^t |x_0 - x^*(b)|$. ■

3. Two-Dimensional Coupled System

We extend g_b to a coupled system on $\mathbb{R}^2$:

$$G_{b,\alpha}(x, y) = (g_b(x) + \alpha y, g_b(y) + \alpha x).$$

Theorem 3.1 (Contraction for small coupling).

For $|\alpha| < 1/2$, the map $G_{b,\alpha}$ is a contraction on $\mathbb{R}^2$ with the $\blacksquare^1$ norm, with Lipschitz constant $L = 1/2 + |\alpha| < 1$.

Proof. For $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$:

$$\begin{aligned} & \mathbf{G}_{b,\alpha}(x_1, y_1) - \mathbf{G}_{b,\alpha}(x_2, y_2) \\ & \leq |g_b(x_1) - g_b(x_2)| + |\alpha||y_1 - y_2| + |g_b(y_1) - g_b(y_2)| + |\alpha||x_1 - x_2| \\ & \leq (1/2 + |\alpha|)(|x_1 - x_2| + |y_1 - y_2|), \text{ using Lemma 2.2 and the triangle inequality. For } |\alpha| < 1/2, \text{ the constant } 1/2 + |\alpha| < 1. \blacksquare \end{aligned}$$

Corollary 3.2.

For $|\alpha| < 1/2$, $G_{b,\alpha}$ has a unique fixed point in $\mathbb{R}^2$, and every orbit converges to it geometrically.

Proof. Immediate from Theorem 3.1 and the Banach fixed-point theorem, since $\mathbb{R}^2$ is complete. $\blacksquare$

4. Saturated Pitchfork Bifurcation

At the critical coupling $|\alpha| = 1/2$ the system undergoes a pitchfork bifurcation. We establish this via a centre-manifold argument.

Theorem 4.1 (Saturated pitchfork bifurcation).

At $\alpha = 1/2$ and $b = 0$, the Jacobian of $G_{0,\alpha}$ at the origin has eigenvalues $\{0, 1\}$. A centre-manifold reduction yields the normal-form pitchfork bifurcation $x \mapsto x - cx^3$ ($c > 0$) on the centre manifold. All branches remain bounded in $[-1, 1]^2$ due to the clipping function κ .

Proof. The Jacobian at the origin is $J = [[1/2, \alpha], [\alpha, 1/2]]$. Eigenvalues are $\lambda_{\pm} = 1/2 \pm \alpha$. At $\alpha = 1/2$: $\lambda_+ = 1$ (centre direction), $\lambda_- = 0$ (stable direction). The centre manifold theorem applies to the $\lambda_+ = 1$ eigendirection. Taylor expanding κ near $x = 0$: $\kappa(x) = x - x^3/3 + O(x^5)$. The reduced dynamics on the centre manifold take the form $x \mapsto x - (1/6)x^3 + O(x^5)$, which is the standard subcritical pitchfork normal form with $c = 1/6 > 0$. Boundedness follows because κ clips all values to $[-1, 1]$. $\blacksquare$

5. Algebraic Separation Theorem

We consider the linearized operator algebra on a real vector space of dimension n . Let $M \in \mathbb{R}^{n \times n}$ with spectral radius $\rho(M) = \max_i |\lambda_i| = 1$.

Theorem 5.1 (Trace bound).

For any $M \in \mathbb{R}^{n \times n}$ with $\rho(M) = 1$ and any $k \in \mathbb{N}$: $|\text{Tr}(M^k)| \leq n$.

Proof. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of M with multiplicity. Since $\rho(M) = 1$, we have $|\lambda_i| \leq 1$ for all i , so $|\lambda_i^k| \leq 1$. Then $|\text{Tr}(M^k)| = |\sum_i \lambda_i^k| \leq \sum_i |\lambda_i^k| \leq n$. $\blacksquare$

Corollary 5.2.

No matrix $M \in \mathbb{R}^{n \times n}$ with $\rho(M) = 1$ and $n < 33$ can satisfy $\text{Tr}(M^k) = 33$ for any $k \in \mathbb{N}$.

Remark. This theorem concerns the linearized algebraic layer only. It makes no claim about persistent-homology invariants of the nonlinear pipeline defined in Section 1; that question is open.

6. Lean 4 Formal Nucleus

The following Lean 4 code formalizes Definitions 1.2, 2.1, and Theorem 2.1. The contraction lemma is stated with a sorry pending identification of the appropriate Mathlib lemma; the bound follows directly from Lemma 2.2.

```
-- Definitions
def kappa (x : Real) : Real := max (-1) (min 1 x)
def g_b (b x : Real) : Real := 0.5 * kappa x + b

-- Fixed point
def x_star (b : Real) : Real :=
  if b < -0.5 then b - 0.5
  else if b <= 0.5 then 2 * b
  else b + 0.5

-- Theorem 2.1
theorem x_star_fixed (b : Real) :
  g_b b (x_star b) = x_star b := by
  simp [g_b, x_star, kappa]
  split_ifs <;> norm_num <;> linarith

-- Lemma 2.2 (sorry pending Mathlib clip lemma)
theorem g_b_contraction (b : Real) :
  LipschitzWith (0.5 : NNReal) (g_b b) := by
  sorry
```

7. Open Questions

The following questions are explicitly open and not addressed in this paper.

Q1. Does there exist an initial condition u_0 and parameter settings (τ, b) such that the Betti sum $\beta_0 + \beta_1$ of $G^k(u_0)$ stabilizes at a value greater than 1 for large k ? Computational experiments with the operators of Section 1 have produced Betti sum = 1; whether a nontrivial invariant exists for some parameter regime is unknown.

Q2. Can the Comp operator C be redesigned — while remaining explicitly defined — so that the pipeline generates and preserves topological features across iterations?

Q3. Close the sorry in the Lean nucleus by identifying the Mathlib lemma for the clipped-linear Lipschitz bound.

8. Discussion

The four operators of the pipeline $G = U \blacksquare F \blacksquare K \blacksquare C$ are defined by explicit pointwise formulas (Section 1). Under this definition the one-dimensional system g_b is a contraction with rate $1/2$, its fixed point is unique and given in closed form, and every orbit converges geometrically (Theorems 2.1–2.3). The two-dimensional coupled extension contracts for $|\alpha| < 1/2$ (Theorem 3.1) and undergoes a saturated pitchfork bifurcation at $|\alpha| = 1/2$ (Theorem 4.1). The algebraic separation theorem (Theorem 5.1) is a standard eigenvalue estimate included for completeness.

No cross-domain application, persistent-homology invariant, or numerical constant beyond those proved above is claimed in this paper. The behaviour of the full 2D discrete pipeline on finite grids is an open computational question addressed in ongoing work.